

ARCHIT SHUKLA

EDUCATION

San Francisco Bay Area, CA | +1 303 931 8281 | architshukla546@gmail.com | Portfolio

University Of Colorado Boulder • Boulder, CO
Masters of Science • Computer Science • GPA: 3.9/4

Aug 2024 - May 2026

Relevant Coursework: Generative AI, Machine Learning, Computer Vision, Advanced Algorithms

SRM Institute of Science & Technology • Chennai, India

Aug 2020 - May 2024

Bachelor of Technology • Computer Science & Engineering • GPA: 3.81/4

WORK EXPERIENCE

University of Colorado Boulder

Jan 2026 – May 2026

Graduate Teaching Assistant—Lab Systems

Boulder, CO

- Led **3 lab sections (55+ students)**, resolving application, networking, and system issues to ensure reliable lab operations.
- Analyzed workflow data using **R** to identify bottlenecks and drive process improvements across lab support and instruction.

Peleg Lab, BioFrontiers Institute

Aug 2025 – Jan 2026

Graduate Researcher-Distributed Systems Engineering

Boulder, CO

- Increased evaluation throughput by **35%** and eliminated hundreds of GPU-hours per cycle by architecting **parallel AI pipelines in Python/C++ on GCP**, enabling scalable distributed experimentation.
- Built **containerized GPU-backed environments** to improve reproducibility, orchestration, & reliability for large-scale AI workloads.
- Reduced failure diagnosis time by **40%** through **GDB/Valgrind debugging and performance profiling**, improving fault isolation and system stability.

CodeClause

June 2024 – July 2024

Software Engineering Intern-Backend & Data Systems

Bangalore, India

- Increased backend throughput by **20%** and reduced API latency by optimizing **Flask REST services, SQL queries, and request workflows**, supporting thousands of daily requests across concurrent workloads.
- Reduced release failures by **25%** through enhancing **CI/CD pipelines** with Jenkins, GitHub, and containerized deployments, improving reliability across backend services.
- Decreased issue diagnosis time by **35%** through implementing **unit testing, structured logging, monitoring, and API debugging**, strengthening observability and production stability.

Adflix Media

Apr 2023 – Sep 2023

Software Engineer Intern-Big Data Infrastructure

Pune, India

- Processed **5M+ records daily** by scaling **SQL-driven ETL pipelines and Dockerized microservices**, improving throughput, reliability, and service communication across distributed cloud workloads.
- Reduced issue detection time by **30%** through building **Grafana dashboards** for latency, throughput, infrastructure, and network monitoring, strengthening observability across production systems.
- Cut recovery time by **25%** by implementing **structured logging, cloud monitoring, automated alerting, and access-controlled operational workflows**, improving fault detection, visibility, resilience, and operational security.

PROJECTS

Multi-Agent Relationship Intelligence Platform | LangGraph, RAG & Knowledge Graphs

Jun 2026 – Present

- Built a **multimodal relationship intelligence platform** that automatically extracted, enriched, and organized contacts from emails, documents, business cards, and meeting notes, maintaining a dynamic **knowledge graph of 1,000+ entities**.
- Designed **LangGraph-based multi-agent workflows** for contact extraction, relationship mapping, follow-up generation, and contextual retrieval, reducing manual networking effort by **80%** through automated evaluations and feedback loops.
- Engineered **JWT-secured FastAPI services** backed by PostgreSQL, Neo4j, and Qdrant, implementing **RAG, context engineering, vector search, and agent observability** pipelines deployed with Docker and AWS.

RepoGraph V2 – Distributed Debugging Platform | Backend Services + Static Analysis [Paper]

Aug 2025 – Dec 2025

- Built a **distributed Python backend** on Linux/UNIX to localize and verify flaky tests across **100K+ LOC repositories** using graph-based code analysis.
- Architected **code intelligence pipelines and REST APIs**, enabling scalable fault localization and automated patch verification.
- Reduced inter-service data transfer by **70–85%** through **graph-scoped retrieval and optimized serialization**, improving throughput and stability under high-concurrency workloads.

LoanSphere – Backend Data Processing Platform | APIs + Data Pipelines [Demo]

Jan 2025 – May 2025

- Built **SQL-driven data pipelines** processing **17M+ records**, implementing scalable cleaning, normalization, and feature engineering across high-volume datasets.
- Developed **FastAPI, Flask, and Node.js microservices** with versioned **REST APIs**, reducing validation errors by **30%** while supporting concurrent requests.
- Automated deployments through **Dockerized CI/CD pipelines**, improving reproducibility, consistency, and release speed.

TECHNICAL SKILLS

- **Languages:** Python, JavaScript, TypeScript, SQL, Java, C++, Go
- **AI Engineering:** Multi-Agent Workflows, RAG Pipelines, LangChain, LangGraph, Context Engineering, Vector Search, Embeddings, LLM Evaluation, Fine-Tuning, AI Observability, Token Optimization, Multimodal AI Applications
- **Backend & Systems:** FastAPI, Node.js, Express.js, REST APIs, Microservices, Distributed Systems, Async & Event-Driven Architectures, JWT, OAuth 2.0, RBAC, Linux
- **Databases & Data:** PostgreSQL, MongoDB, Redis, DynamoDB, ETL Pipelines, Query Optimization, Vector Databases (Qdrant)
- **Cloud & DevOps:** AWS (EC2, S3, IAM, ECS), GCP, Docker, Kubernetes, Terraform, CI/CD, GitHub Actions, Linux, Git, Nginx
- **Networking & Security:** DNS, HTTP/HTTPS, TCP/IP, TLS/SSL, Load Balancing, CDN, Reverse Proxies, Secure API Design, OWASP Fundamentals, Rate Limiting
- **Testing & Reliability:** Pytest, Jest, Playwright, Unit, Integration & End-to-End Testing, OpenTelemetry, Observability